

Lecture 2 – Energy Basics

Dr. Yu Zhang

ECE Department, UC Santa Cruz

ECE180J



Outline

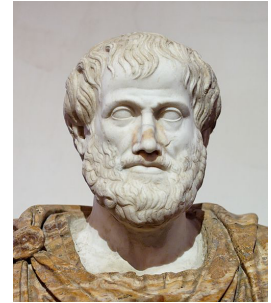
- Energy and Power
- Law of Conservation of Energy
- Household Appliance Power Ratings

B1: *Renewable and Efficient Electric Power Systems* (2nd ed): Chap 1 & 2

B3: *Renewable Energy Power for a Sustainable Future* (4th ed): Chap 1, Appendix A

Energy

- The word energy derives from the Greek *en* (in) and *ergon* (work) [*wiki: Ancient Greek: ἐνέργεια, energeia, meaning 'activity, operation', appeared for the 1st time in Aristotle's work in the 4th century BC*]
- In 1807, Thomas Young was possibly the first to use the term “energy” in its modern sense.
- Broad definition: **the capacity doing work**
How much potential a physical system has to change

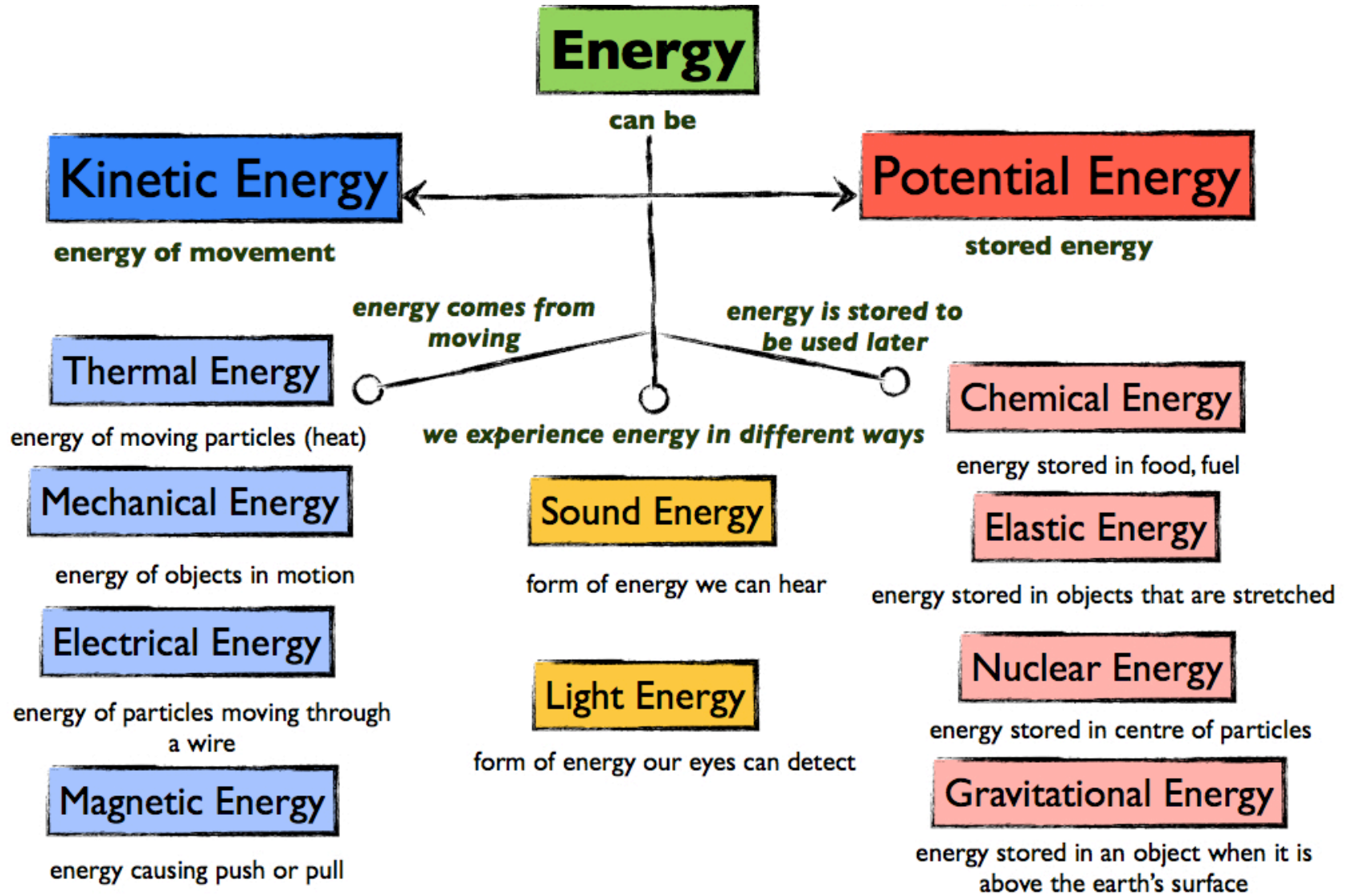


Aristotle, Greek philosopher and polymath



Thomas Young, British polymath and physician

Forms of Energy

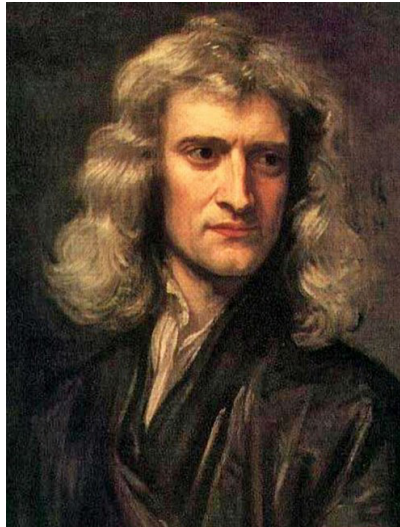


Energy Type	Example	How Energy Does Useful Work
Mechanical Energy	Riding a bicycle	Muscles convert chemical energy into motion to move the bike forward.
Electrical Energy	Charging a smartphone	Electrical energy from the charger powers the phone's battery.
Thermal Energy	Cooking on a stove	Heat energy cooks food by raising its temperature.
Chemical Energy	Eating food	The body converts food's chemical energy into movement and heat.
Nuclear Energy	Sunlight warming the Earth	Nuclear fusion in the Sun provides heat and light, enabling life.
Kinetic Energy	Driving a car	Fuel combustion powers the engine, moving the vehicle.
Potential Energy	Water in a dam	Stored water releases energy when it flows down, generating electricity.
Solar Energy	Using a solar calculator	Sunlight is converted into electricity to power the device.
Wind Energy	Flying a kite	Wind's kinetic energy lifts the kite into the air.

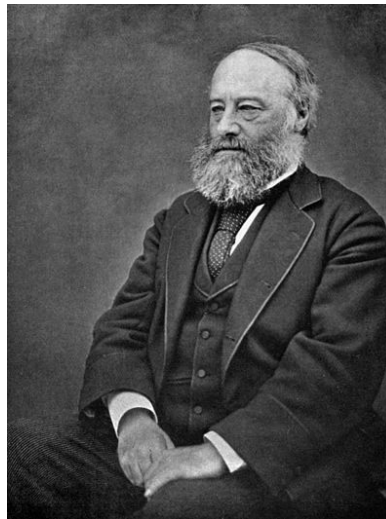
Unit of Energy

The energy supplied by force of 1 Newton in causing movement through 1 meter.

- Energy (J) = Force (N) x Distance (m)
- Unit of energy: Joule (J)



Isaac Newton (1642-1727)
English physicist



James Joule (1818-1889)
English physicist

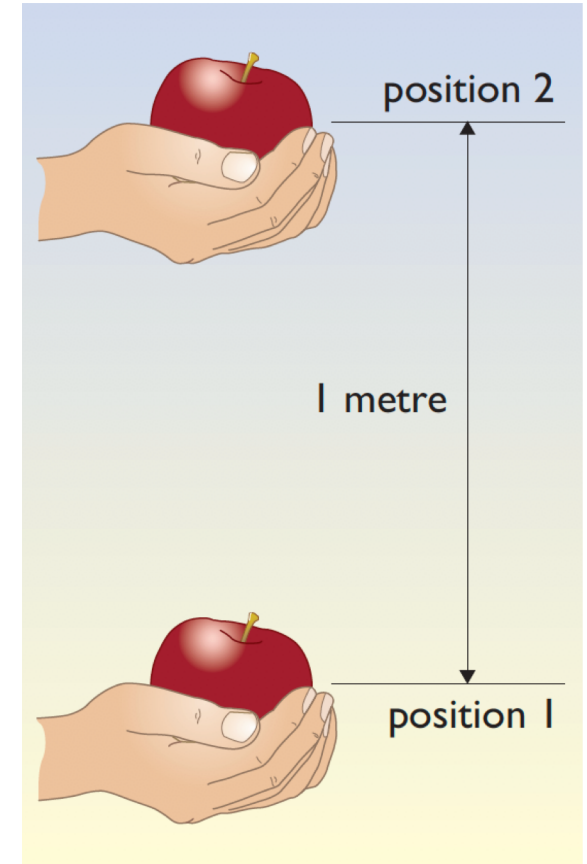


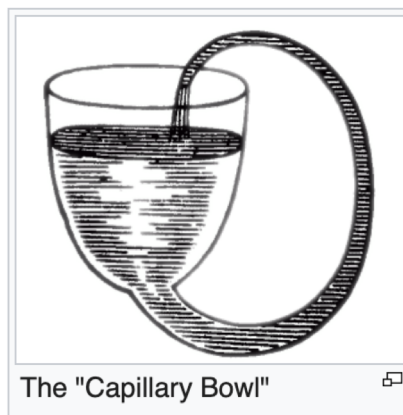
Figure 1.1 The amount of energy required to raise a 100 g apple vertically through 1 m is approximately one joule (1 J)

Conservation of Energy

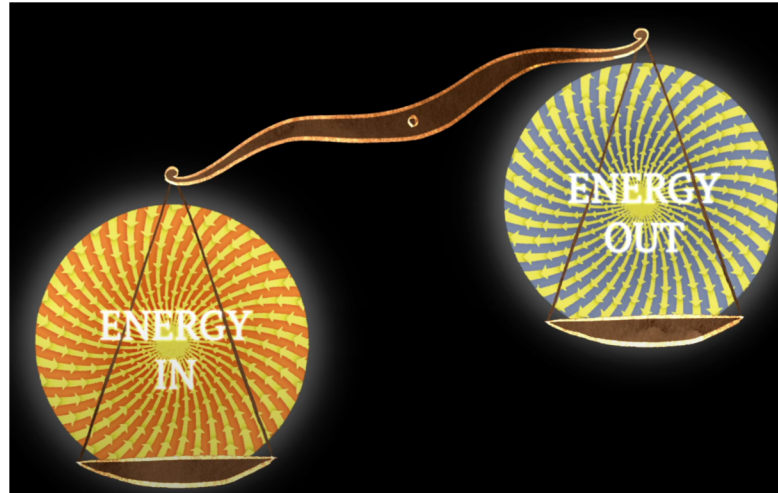
Law of Conservation of Energy:

Energy is a property that is not created/destroyed, although energy can change in form.

- **First law of thermodynamics:** The total energy of an isolated system remains constant (be conserved over time).
- A perpetual motion machine of the first kind cannot exist: no system without an external energy supply can deliver an unlimited amount of energy to its surroundings.



Perpetual Motion Machine!?



Three Laws of Thermodynamics:

- **The 1st law:** Energy can neither be created nor destroyed.
- **The 2nd law:** The entropy of any isolated system always increases.
- **The 3rd law:** The entropy of a system approaches a constant value as the temperature approaches absolute zero.

Conservation of energy can arguably be violated by general relativity on the cosmological scale.

Einstein tells us that space and time are dynamical, and in particular that they can evolve with time. When the space through which particles move is changing, the total energy of those particles is not conserved.

Noether's Theorem

Noether's Theorem: For every continuous global symmetry, there exists a conserved quantity.

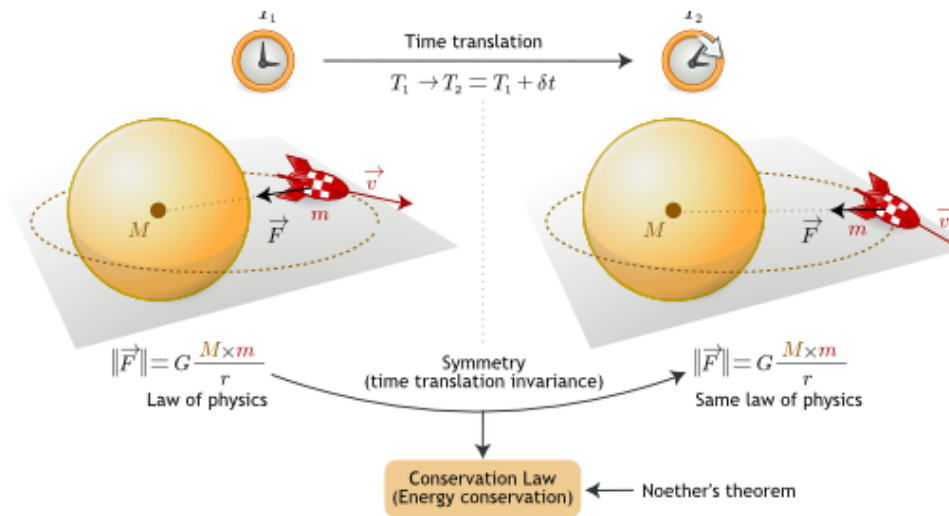
- Global means that the symmetry must uniformly apply to the entire system and not vary spatially across the system.
- Continuous means the symmetry can be applied for a continuous range of some parameter, such as a shift or a rotation.



Emmy Noether (1882-1935)
German mathematician

Symmetry	Conservation law
Time-translation	Energy
Space-translation	Momentum
Rotation	Angular momentum

Time Symmetry and Energy Conservation



Time-translation symmetry: the laws of physics do not change over time (time invariance).

Fig source: quantum-bits

If physics really does change over time; e.g., gravitation constant G .

- Then, we can always raise the weight when gravity is weak and lower it when gravity is strong.
- The key: the energy we get back out is greater than what we originally put in. Each time, we make a profit in energy.
- We keep running the cycle and become richer and richer.

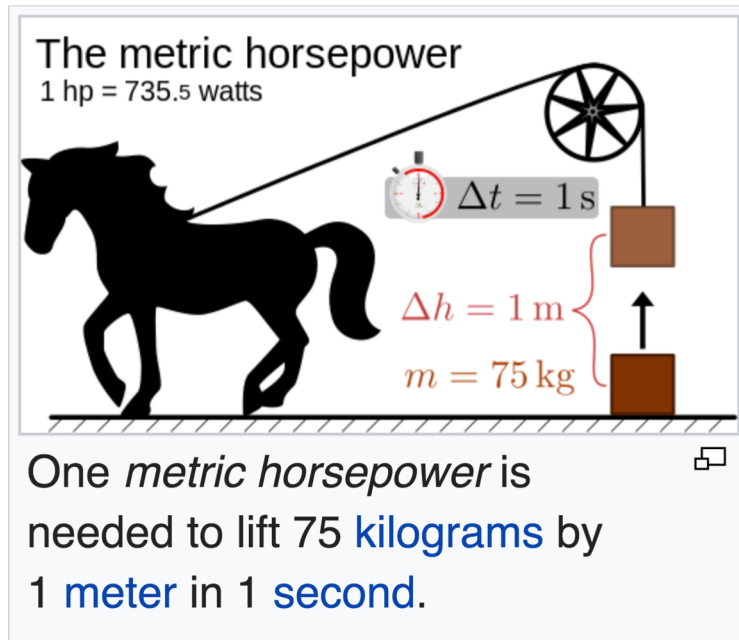
Hence, conservation in energy implies time symmetry.

Noether's Theorem asserts that it works the other way around as well, which requires in-depth mathematical derivations with Lagrangian function and principle of least action.

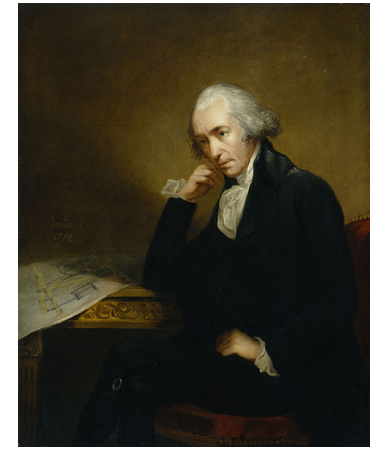
Power

➤ Power is the amount of energy transferred or converted per unit time (rate of doing work)

- Power (W) = Energy (J) / Time (sec)
- Unit of power: Watt (W)



$$1\text{hp} = \frac{mg \times d}{t} = \frac{75 \times 9.8 \times 1}{1\text{s}} = 735\text{W}$$



James Watt (1736-1819)
Scottish inventor



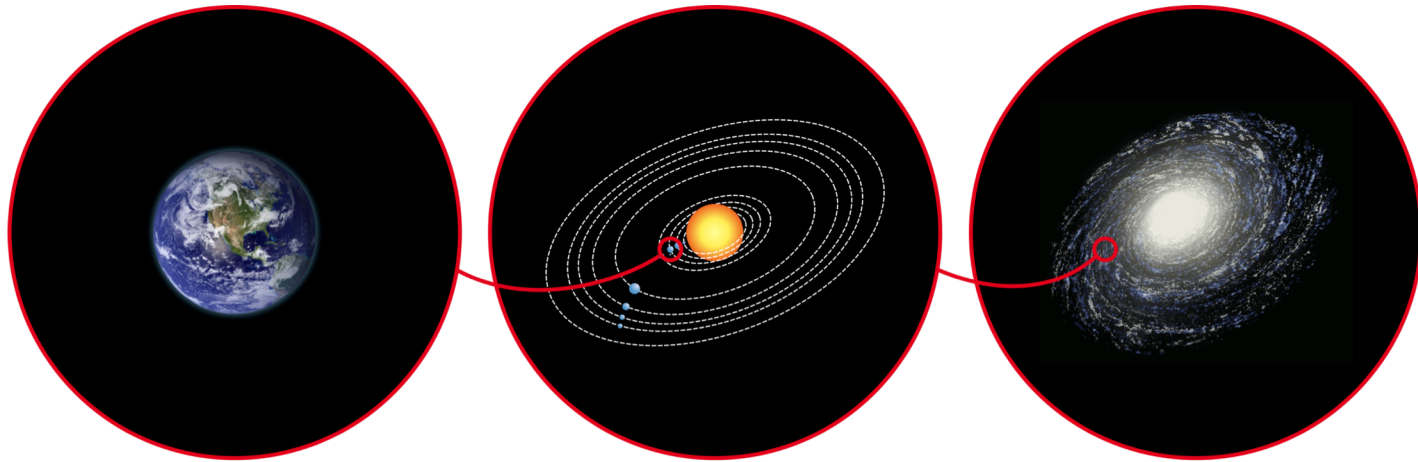
Toyota Camry SE: 203 hp



Porsche 911 GT2: 700 hp

Civilization's Level Described by Power

In 1964, Soviet astronomer Nikolai Kardashev defined **three levels of civilization**, based on the order of magnitude of power available to each:



Type I: 10^{16} W

Type II: 10^{26} W

Type III: 10^{36} W

- Type I (**planetary civilization**): can use and store all of the energy available on its planet.
- Type II (**stellar civilization**): can use and control energy at the scale of its planetary system.
- Type III (**galactic civilization**): can control energy at the scale of its entire host galaxy.

Mechanical Power

Power in mechanical systems is the combination of forces and movement.

- The product of a force on an object and the object's velocity $P = \mathbf{F} \cdot \mathbf{v}$
- The product of a torque on a shaft and the shaft's angular velocity $P = \tau \cdot \omega$

Assume a constant force $\mathbf{F}$ throughout a distance $\mathbf{x}$, then

$$P = \frac{dW}{dt} = \frac{d}{dt} (\mathbf{F} \cdot \mathbf{x}) = \mathbf{F} \cdot \frac{d\mathbf{x}}{dt} = \mathbf{F} \cdot \mathbf{v}.$$

If we have a variable force $\mathbf{F}$ over a curve C : then

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_{\Delta t} \mathbf{F} \cdot \frac{d\mathbf{r}}{dt} dt = \int_{\Delta t} \mathbf{F} \cdot \mathbf{v} dt.$$



Fundamental theorem
of calculus

$$P = \frac{dW}{dt} = \frac{d}{dt} \int_{\Delta t} \mathbf{F} \cdot \mathbf{v} dt = \mathbf{F} \cdot \mathbf{v}.$$

International System of Units (SI)

Base quantity	SI base unit	
	Name	Symbol
length	meter	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
thermodynamic temp	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd



SI-derived Units

SI derived units are formed by powers, products or quotients of the base units

Name ^{note 1}	Symbol	Quantity	In other SI units	In SI base units
radian ^{note 2}	rad	angle		$(\text{m} \cdot \text{m}^{-1})$
steradian ^{note 2}	sr	solid angle		$(\text{m}^2 \cdot \text{m}^{-2})$
hertz	Hz	frequency		s^{-1}
newton	N	force, weight		$\text{kg} \cdot \text{m} \cdot \text{s}^{-2}$
pascal	Pa	pressure, stress	N/m^2	$\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2}$
joule	J	energy, work, heat	$\text{N} \cdot \text{m}$	$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$
watt	W	power, radiant flux	J/s	$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3}$
coulomb	C	electric charge or quantity of electricity		$\text{s} \cdot \text{A}$
volt	V	voltage (electrical potential), emf	W/A	$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3} \cdot \text{A}^{-1}$

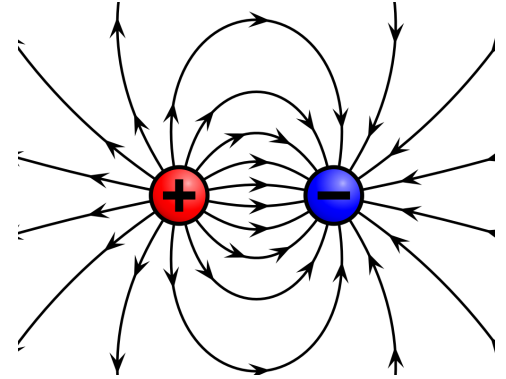
An example (change of units):

$$\begin{aligned} \text{J} &= \text{N} \cdot \text{m} = \text{Pa} \cdot \text{m}^3 = \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} \\ &= \text{W} \cdot \text{s} = \text{C} \cdot \text{V} \end{aligned}$$

Electric Charges

Charge (q): a fundamental property of matter that gives rise to electromagnetic interactions.

- an intrinsic property of particles such as electrons (negative charged) and protons (positive charged).
- particles with the same type of charge repel each other, while particles with opposite charges attract each other.



Quantization of charge: Charge is quantized (occurs in discrete packets). The smallest unit of charge is the charge of an electron (or proton), which is approximately 1.6×10^{-19} coulombs (C).

Electric fields: Charges create electric fields around them, which exert forces on other charges placed in the field.

- *Coulomb's Law:* This law describes the electrostatic force between two charged particles. The force is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.

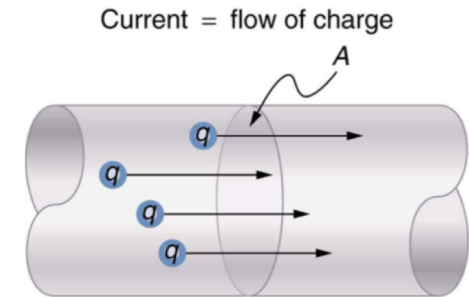
$$F = k \frac{q_1 q_2}{r^2}$$

The behavior of charges in matter is fundamental to understanding phenomena like electricity, magnetism, and electromagnetic radiation.

Current and Voltage

Current (i): Movement of free electrons. Defined as the rate of flow of electric charge through a cross-sectional area of a conductor per unit time.

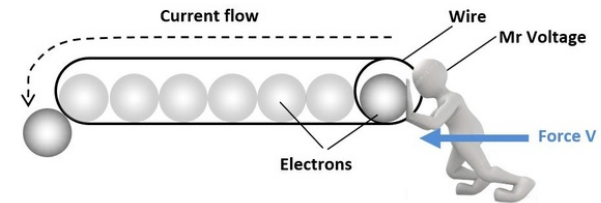
$$i := \frac{dq}{dt}$$



Direction of current: By convention, the direction of current flow is considered to be from the positive (+) to the negative (-) terminal of a battery or power source. This is known as conventional current flow. Electrons flow in the opposite direction (from negative to positive)

Voltage (v): Electric potential difference between two points, which is defined as the amount of energy (w) in moving a unit of charge (q) from one point to another.

$$v := \frac{dw}{dq}$$



- Voltage is the work done (energy transferred) per unit charge when moving a positive test charge between two points in an electric field.
- Electric charge flows from points of higher voltage to points of lower voltage.

Quiz

By the definitions of power, voltage and current, derive their relationship: $P=VI$

$$p = \frac{dw}{dt} = \frac{dw}{dq} \times \frac{dq}{dt} = v \times i$$

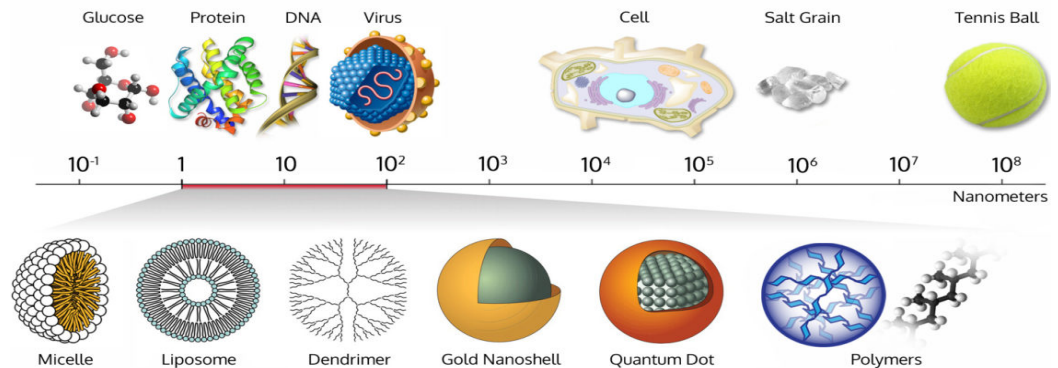
Relationships of Key Quantities

Electrical Quantity	Symbol	Unit	Abbreviation	Relationship
Charge	q	coulomb	C	$q = \int i \, dt$
Current	i	ampere	A	$i = dq/dt$
Voltage	v	volt	V	$v = dw/dq$
Power	p	joule/second or watt	J/s W	$p = dw/dt$
Energy	w	joule or watt-hour	J Wh	$w = \int p \, dt$

Scientific Notation

Small Quantities			Large Quantities		
Quantity	Prefix	Symbol	Quantity	Prefix	Symbol
10^{-3}	milli	m	10^3	kilo	k
10^{-6}	micro	μ	10^6	mega	M
10^{-9}	nano	n	10^9	giga	G
10^{-12}	pico	p	10^{12}	tera	T

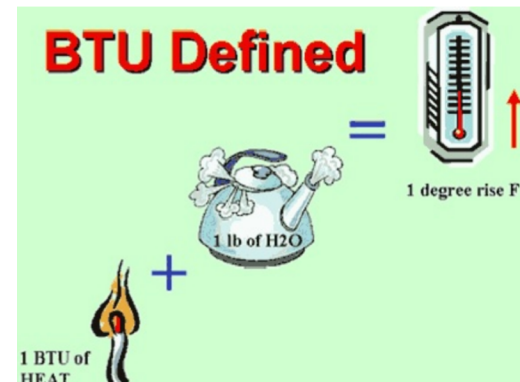
The IBM Model 350 disk file with a storage space of 5MB from 1956 and a Micro SD Card



See more: [The Smallest to the Biggest Thing in the Universe!](#)

Popular Units for Comparing Energy

- **British thermal units (Btu)** = the amount of heat to raise the temperature of one pound of water by one degree Fahrenheit.
- barrels of oil equivalent
- metric tons of oil (coal) equivalent



- 1 kWh of electricity = 3,412 Btu
- 1 cubic foot of natural gas = 1,037 Btu
- 1 gallon of heating oil = 138,500 Btu
- 1 barrel of crude oil = 5,800,000 Btu
- 1 gallon of gasoline = 114,100 Btu
- 1 gallon of diesel fuel = 137,452 Btu
- 1 gallon of propane = 91,333 Btu
- 1 short ton of coal = 24,000,000 Btu (typical average)

The unit of million Btu (mmBtu) is more commonly used

Converting to Btu

Example 1:

- Your home has a natural gas furnace that used 81,300 cubic feet of natural gas for heating last winter.
- Your neighbor has a furnace that burns heating oil, which used 584 gallons of heating oil last winter.

Q: Which home used more energy for heating?

Natural gas: 81,300 cubic feet x 1,037 Btu/cubic foot = 84.3 mmBtu

Heating oil: 584 gallons x 138,500 Btu/gallon = 80.9 mmBtu

A: You used more energy to heat your home!

Converting to Btu (cont'd)

Example 2:

You need a new furnace. You consider heating systems that use natural gas or heating oil. The costs of the fuels: \$10.40 per thousand cubic feet (natural gas), and \$2.70 per gallon (heating oil).

Q: Which one to buy?

Solution: Compare the price of the fuels on an equal basis

Natural gas	\$10.40 per thousand cubic feet	÷	1.037 million Btu per thousand cubic feet	= \$10.03 per million Btu
Heating oil	\$2.70 per gallon	÷	0.1385 million Btu per gallon	= \$19.49 per million Btu

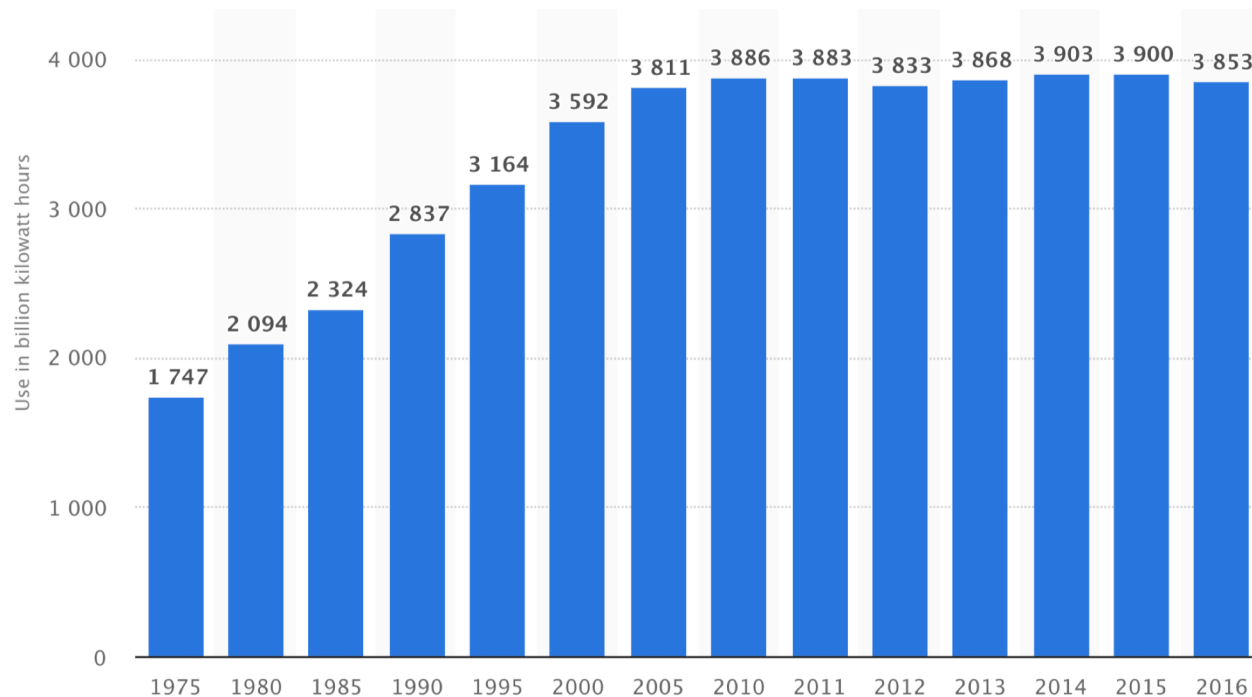
A: Go for natural gas one

Energy Consumption



Ex 1: A cell phone camera consumes 10^{-8} Joules/pixel for A/D conversion
1080x1920 pixels/frame = 2.1million pixels/frame = 0.021 J/frame
30 frames/second for video = 0.62 J/sec = 0.62 watts.

Ex 2: U.S. annual electric energy consumption is about 3.9 trillion kWh
(about 11,974 kWh per person, which means on average we each use
1.37 kW of power continuously).

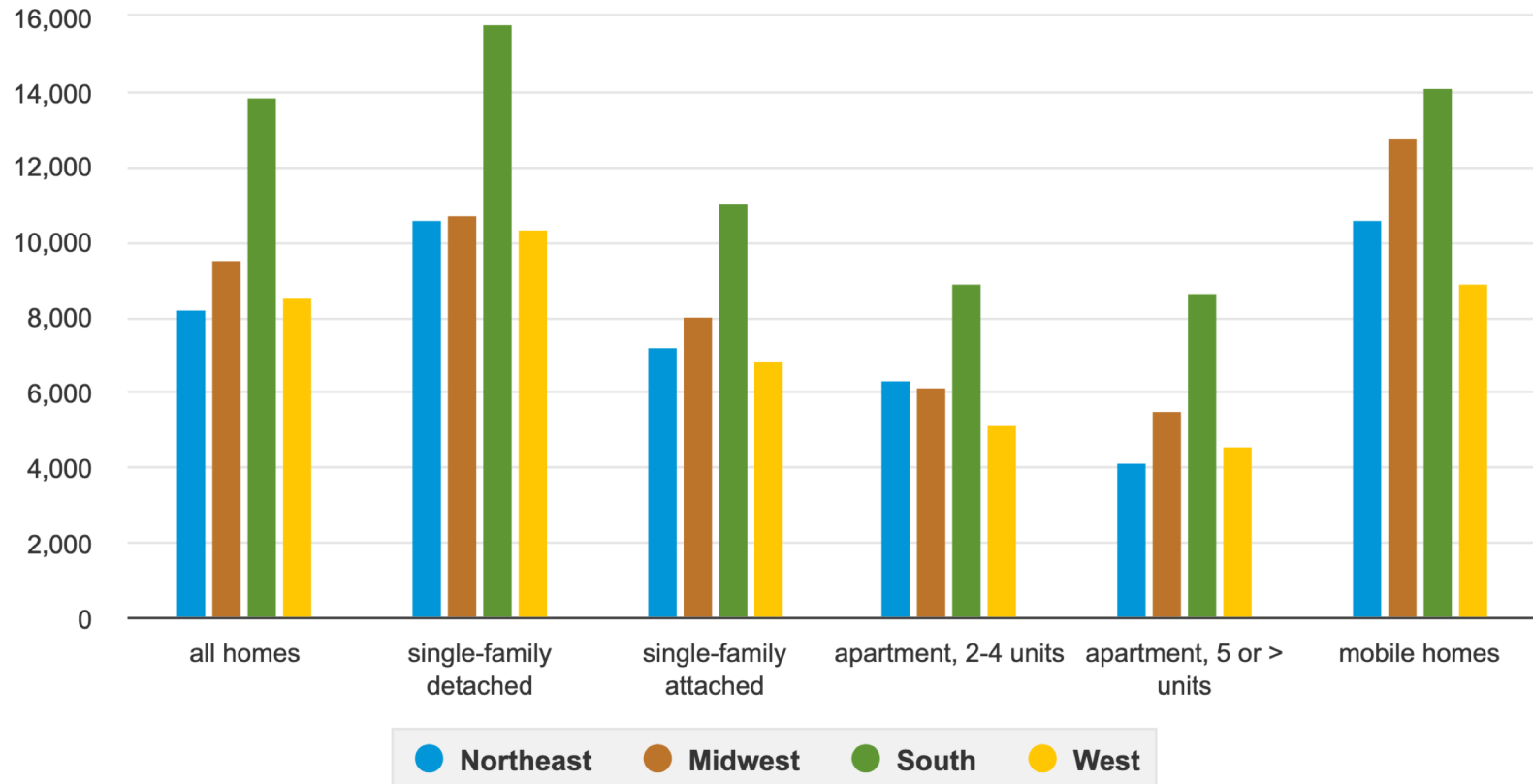


Household Annual Energy Consumption

Average annual electricity consumption by type of home and census region, 2015



kilowatthours

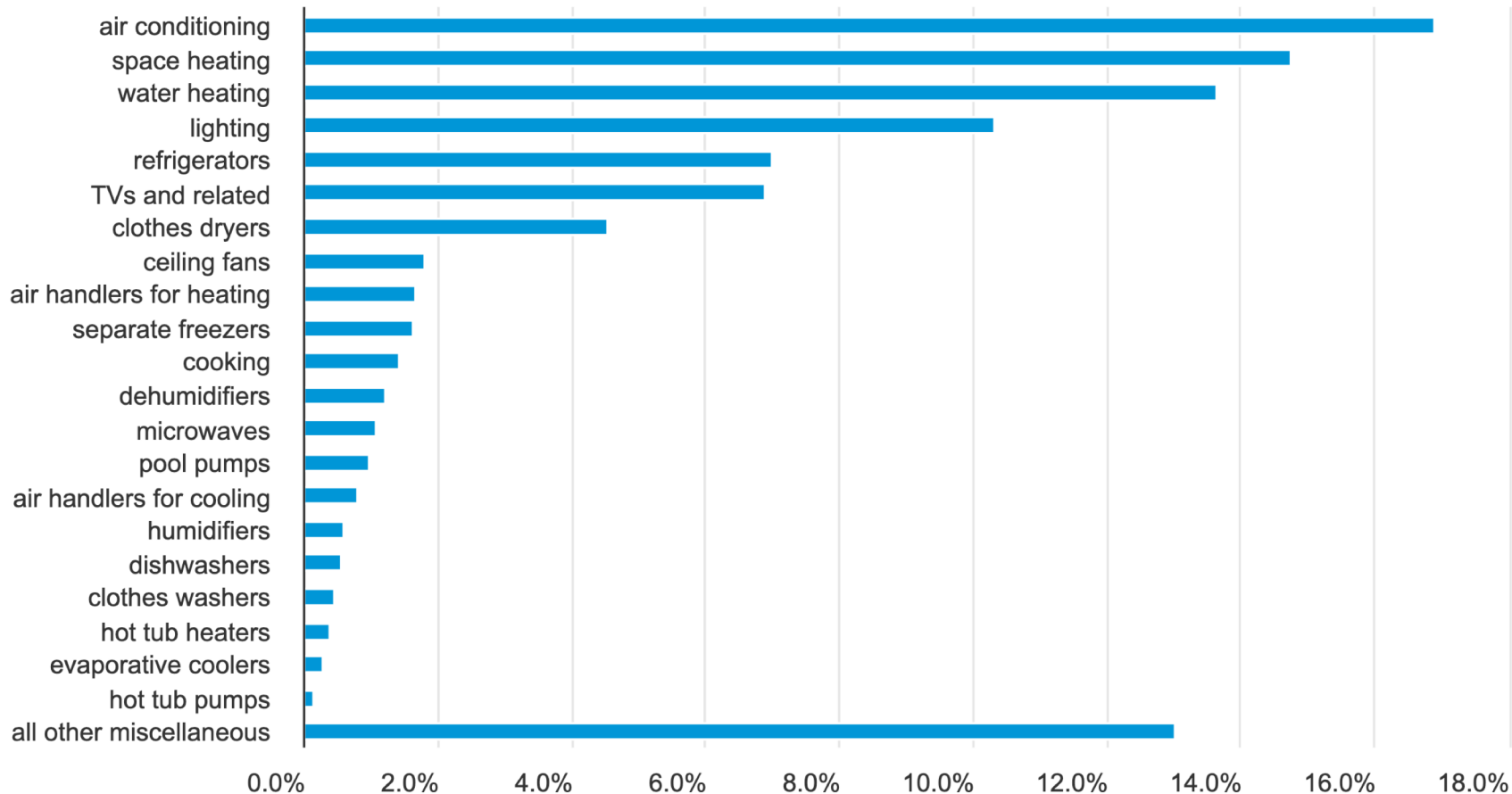


Source: U.S. Energy Information Administration, 2015 Residential Energy Consumption Survey

Energy Consumption Percentage of End Use

Residential site electricity consumption by end use, 2015

percent of total



Source: U.S. Energy Information Administration, 2015 Residential Energy Consumption Survey

Typical Power Ratings of Appliances

ESSENTIAL APPLIANCES

ESTIMATED WATTAGE



Household Appliances ▲	Rated (Running) Watts ⚡	Additional Surge Watts ⚡
Ceiling Fan	60 W	70 W
Central AC (10,000 BTU)	1,500 W	4,500 W
Central AC (24,000 BTU)	3,800 W	11,400 W
Central AC (40,000 BTU)	6,000 W	6,700 W
Dehumidifier	240 W	0 W
Electric Heater (Fan)	2,000 W	1,000 W
Electric Thermal Radiator	500 W	0 W
Electric Water Heater	4,000 W	0 W
Electric Water Heater (Immersion)	3,000 W	0 W
Electric Water Heater (Tankless)	6,600 W	2,200 W

KITCHEN APPLIANCES

ESTIMATED WATTAGE



Household Appliances ▲	Rated (Running) Watts ⚡	Additional Surge Watts ⚡
Air Fryer	1,500 W	0 W
Coffee Maker	1,000 W	0 W
Cooker Hood	20 W	10 W
Deep Freezer	500 W	1,500 W
Dishwasher	1,500 W	1,500 W
Electric Can Opener	170 W	0 W
Electric Kettle	1,200 W	3,000 W
Electric Oven	2,150 W	0 W
Electric Stove (8" Element)	2,100 W	0 W
Espresso Coffee Machine	1,300 W	200 W